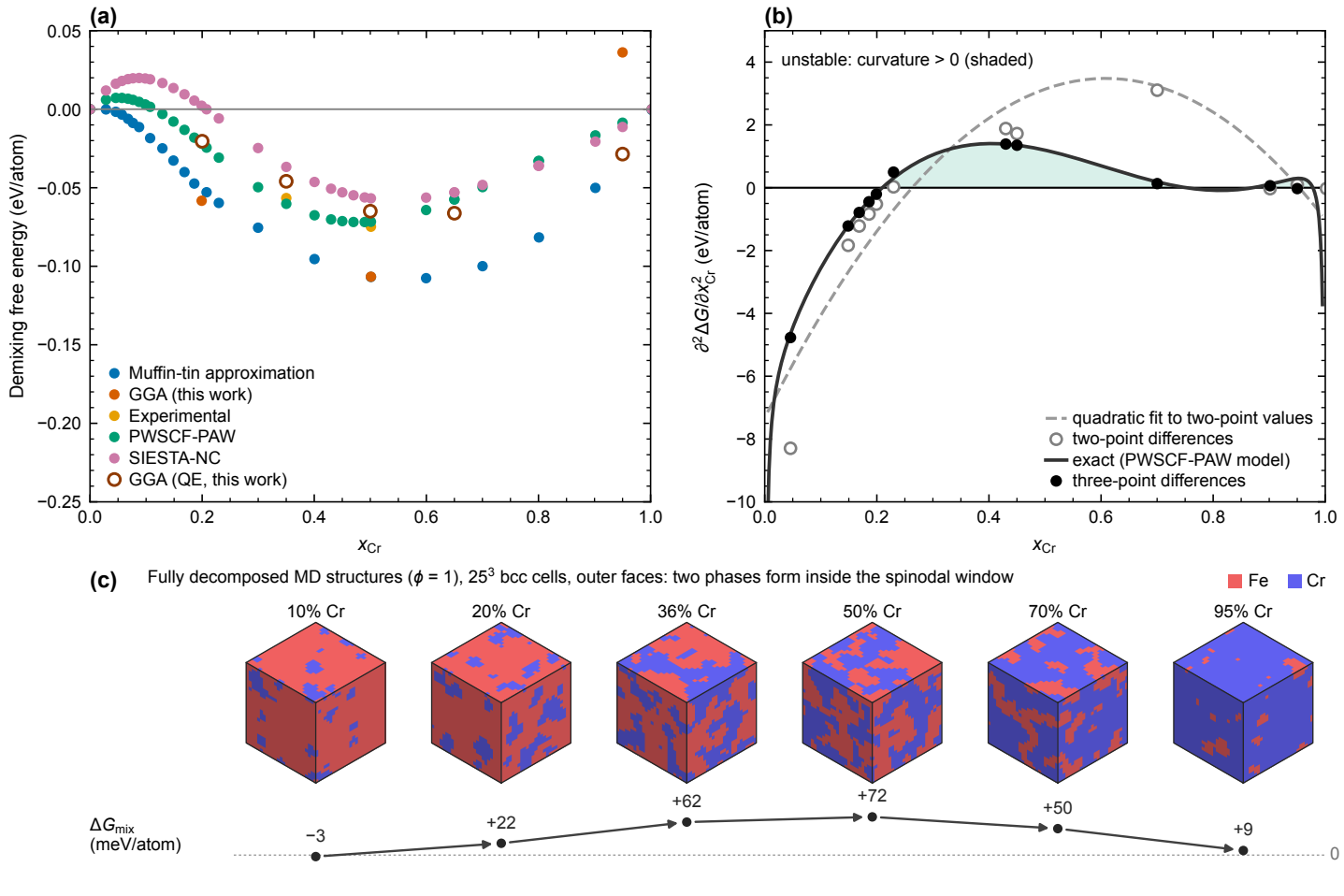

(a)
Demixing free energy (eV/atom)
x_{Cr}
Muffin-tin approximation
GGA (this work)
Experimental
PWSCF-PAW
SIESTA-NC
GGA (QE, this work)
(b)
unstable: curvature > 0 (shaded)
$\partial^2 \Delta G / \partial x_{Cr}^2$ (eV/atom)
x_{Cr}
quadratic fit to two-point values
two-point differences
exact (PWSCF-PAW model)
three-point differences
(c)
Fully decomposed MD structures (ϕ = 1), 25^3 bcc cells, outer faces: two phases form inside the spinodal window
Fe
Cr
10% Cr
20% Cr
36% Cr
50% Cr
70% Cr
95% Cr
ΔG_{mix} (meV/atom)
−3
+22
+62
+72
+50
+9
0